Question

$\mathbf{N}$ is a rare element whose mineral resources are sparsely distributed, often occurring as associated minerals or as impurities in other minerals. Elemental $\mathbf{N}$ is insoluble in water or hydrochloric acid, but can be oxidized to $\mathbf{A}$ by fuming nitric acid; $\mathbf{A}$ is a white amorphous powder, slightly soluble in water, and easily dehydrated; compound $\mathbf{B}$ is obtained after complete dehydration of $\mathbf{A}$. $\mathbf{B}$ reacts with carbon tetrachloride at $723\ \mathrm{K}$ to give a white hygroscopic solid $\mathbf{C}$. $\mathbf{C}$ combines with elemental $\mathbf{N}$ to form $\mathbf{D}$; $\mathbf{D}$ can also be obtained by reacting elemental $\mathbf{N}$ with sulfuryl chloride at $310\mathrm{K}$ in a $1:1$ molar ratio. $\mathbf{D}$ reacts with liquid non-metallic element $\mathbf{E}$ to obtain a yellow solid $\mathbf{F}$, in which the mass fraction of $\mathbf{N}$ is 35.61%. When isolated from air, elemental $\mathbf{N}$ reacts with molten KCN to produce $\mathbf{G}$ and release gas $\mathbf{H}$; $\mathbf{G}$ reacts with acid to produce a highly toxic, foul-smelling gas $\mathbf{I}$; $\mathbf{I}$ spontaneously decomposes to elemental $\mathbf{N}$ at $273\ \mathrm{K}$.

The following options are correct:

A. All other options are incorrect

B. Element $\mathbf{N}$ is a subgroup metal element.

C. Element $\mathbf{N}$ is a main group metal element.

D. The number of binary compounds in $\mathbf{A}-\mathbf{I}$ is 5

E. The number of binary compounds in $\mathbf{A}-\mathbf{I}$ is 7.

F. $\mathbf{B}-\mathbf{I}$ contains 4 substances whose molecular formula contains greater than or equal to 4 atoms.

G. $\mathbf{B}-\mathbf{I}$ contains 4 substances with molecular formulas containing 3 atoms.

Answer

Correct Answer: G

Detailed Explanation

$\mathbf{N}$ is a rare element, insoluble in hydrochloric acid and water, exhibiting properties of an inert metal or a high-period main group nonmetal element;

When $\mathbf{N}$ is fused with KCN, a gas is produced, then the remaining G is basically certain to be a potassium salt containing $\mathbf{N}$; therefore, $\mathbf{N}$ is likely negatively charged, excluding metallic elements.

CHECKPOINT 1 PTS

$\mathbf{N}$ is likely negatively charged, excluding metallic elements

$\mathbf{G}$ reacts with acid to produce the highly toxic gas $\mathbf{I}$, then $\mathbf{I}$ is a hydride of $\mathbf{N}$, containing a highly toxic substance, likely a high-period chalcogen and pnictogen element.

CHECKPOINT 1 PTS

$\mathbf{I}$ is a hydride of $\mathbf{N}$

$\mathbf{I}$ decomposes at room temperature, indicating instability, therefore it can be judged that the nonmetallic property of $\mathbf{N}$ is very weak, belonging to a high period; therefore, it can be basically judged that $\mathbf{N}$ is Te or Se, but H_2Se does not decompose at room temperature, so $\mathbf{N}$ can only be Te. Therefore, options B and C are incorrect.

CHECKPOINT 1 PTS

$\mathbf{N}$'s nonmetallic property is very weak, belonging to a high-period element

CHECKPOINT 2 PTS

Judging $\mathbf{N}$ to be Te based on the properties of the hydride

Te dissolves in fuming nitric acid to produce $\mathbf{A}$, then $\mathbf{A}$ is $\mathrm{TeO_2 \cdot xH_2O}$. The dehydrated product $\mathbf{B}$ is $\mathrm{TeO_2}$; $\mathrm{TeO_2}$ reacts with carbon tetrachloride to obtain a hygroscopic solid $\mathbf{C}$, which can be judged to be a common hydrolyzable chloride $\mathrm{TeCl_4}$. $\mathrm{TeCl_4}$ combines with Te in a comproportionation reaction, and the resulting $\mathbf{D}$ is $\mathrm{TeCl_2}$.

CHECKPOINT 1 PTS

$\mathbf{A}$ is $\mathrm{TeO_2 \cdot xH_2O}$

CHECKPOINT 1 PTS

$\mathbf{B}$ is $\mathrm{TeO_2}$

CHECKPOINT 1 PTS

$\mathbf{C}$ is a common hydrolyzable chloride $TeCl_4$

CHECKPOINT 1 PTS

$\mathbf{D}$ is $TeCl_2$

$TeCl_2$ reacts with the nonmetallic element $\mathbf{E}$ to obtain $\mathbf{F}$, then $\mathbf{F}$ should contain three elements $Te, Cl, \mathbf{E}$. Based on the mass fraction, assuming $\mathbf{F}$ contains one Te, the remaining molecular weight is $127.6/0.3561 - 127.6 = 230.7 = 35.45 \times 2 + 79.9 \times 2$; therefore, $\mathbf{E}$ is Br_2, and $\mathbf{F}$ is $TeCl_2Br_2$.

CHECKPOINT 1 PTS

Inferring that $\mathbf{E}$ is Br_2 based on the mass fraction

CHECKPOINT 1 PTS

$\mathbf{F}$ is $TeCl_2Br_2$

Te reacts with KCN in a fusion reaction to produce gas and potassium salt, then $\mathbf{G}$ is K_2Te, $\mathbf{H}$ is $(CN)_2$, and the highly toxic substance $\mathbf{I}$ is H_2Te.

CHECKPOINT 1 PTS

$\mathbf{G}$ is K_2Te

CHECKPOINT 1 PTS

$\mathbf{H}$ is $(CN)_2$

CHECKPOINT 1 PTS

The highly toxic substance $\mathbf{I}$ is H_2Te

According to the chemical formulas, options D, E, and F are incorrect, and G is correct.